

SOLPS5.0 B2, or B2.5

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- in which we learn how to run B2.5

SOLPSTOP		+---lib
+---src		+--- ...
+---Braams		+---include
+---b2		+---Carre
+---base		+--- ...
+--- ...		+---agg
+---runs		+--- ...
+--- ...		+---aggj
+---B2		+--- ...
+---src_xpb		+---Sonnet
+--- ...		+--- ...
+---src.local		+---doc
+---bin		+---solps
+--- ...		+--- ...

Compiling and running the code:

```
cd solps5.0_MY_ARBITRARY_NAME
source setup.csh
sbb
gmake depend
gmake
cd solps5.0_MY_ARBITRARY_NAME
source setup.csh
sbb
cd ../runs/bench_mark/48x18
mkdir MY_ARBITRARY_DIRECTORY
cd MY_ARBITRARY_DIRECTORY
b2run > run.log
```

Explanation:

- move into the top level directory
- add various aliases, environment variables
- move to the Braams base directory
- setup the dependencies
- and compile
- move to the run directory
- make a new sub-directory
- and move into it
- then run

B2.5 consists of a number of programs

b2ag prepares the geometry and magnetic field

b2ah prepares the default physics parameters

b2ai prepares the initial plasma state

b2ar prepares a table of atomic rate coefficients

b2co one option when running the main program is to write out a short restart file periodically; this program converts such a restart file into the normal version

b2fu the opposite of **b2uf**; it converts a formatted plasma file `b2fplasmf` to the normal unformatted `b2fplasma`

b2mn the main program

b2pl the plot program

b2ts an example post-processing program; it reads the basic output files produced by **b2mn** and prints some simple geometric quantities

b2uf the normal `b2fplasma` file (used for plotting) is written using FORTRAN unformatted writes; this program reads such a file and writes it out as a formatted file, `b2fplasmf`. Typically used for moving the `b2fplasma` to a different computer architecture.

b2xd was used in the transition stage from SOLPS4.0 to SOLPS5.0 — it converts from the SOLPS5.0 plasma state file to that used by SOLPS4.0

b2yg plots the geometry and magnetic field

b2yh prints the table of physics parameters

b2yi plots the plasma state

b2yi_gnuplot a version of b2yi using gnuplot

b2ym claims to produce a movie

b2yn plot the progress of the inner b2 iterations

b2yp displays the plasma state

b2yq claims to provide a quick display of the evolution data

b2yr plot the atomic rate coefficients

b2yt transfers the plasma from one mesh to another mesh of a different size

SOLPSTOP

```

+---src
|   +---Braams
|   |   +---b2
|   |   |   +---base
|   |   |   |   +---config
|   |   |   |   +---IBMaix
|   |   |   |   +---sun5
|   |   |   |   +---unicos
|   |   |   |   +---linux.Fujitsu
|   |   |   +---runs
|   |   |   |   +---bench_mark
|   |   |   |   |   +---48x18
|   |   |   |   |   |   +---baserun
|   |   |   |   |   |   +---ibmtok1
|   |   |   |   |   +---96x36
|   |   |   |   |   |   +---baserun
|   |   |   |   |   |   +---ibmtok1
|   |   |   |   +---192x72
|   |   |   |   |   +---baserun
|   |   |   |   |   +---ibmtok1

```

base used for compiling the code

- config contains configuration files for different compilers
- IBMaix, sun5, unicos, linux.Fujitsu are the .f and .o files for the different compilers/operating systems
- selected by environment variable \$OBJECTCODE which was setup by the source setup.csh in the top of the tree

runs the run directory

- here one global case, bench_mark
- with sub-cases 48x18, 96x36, 192x72
- and within this, a baserun directory and one or more variations, here typified by ibmtok1
- here the different baserun directories contain different geometry information

- when doing a run, the code “looks” for certain information
 - first in the actual run directory
 - and, if not found there, in the baserun directory at the same level
 - (this is done by a Makefile)
- thus the baserun provides a set of defaults that would be shared between a number of cases
 - geometry information
 - default physics information
 - atomic physics rates
 - default initial plasma state
- starting a complete new run therefore means modifying (or at least checking) the following files in baserun
 - b2ag.dat** specifies the grid to be used
 - b2ai.dat** specifies the initial plasma state
 - b2ar.dat** specifies the atomic physics
 - b2ah.dat** the default physics
- and then running the relevant programs with
 - b2run b2ag b2ai b2ar b2ah
- these programs then produce files which are then used by the main program

“dat” & “prt” files

filename	formatted?	read by
b2ag.dat	formatted	b2ag b2ai
b2ah.dat	formatted	b2ah
b2ai.dat	formatted	b2ai
b2ar.dat	formatted	b2ar
b2mn.dat	formatted	b2mn
b2yg.dat	formatted	b2yg
b2yh.dat	formatted	b2yh
b2yi.dat	formatted	b2yi
b2ym.dat	formatted	b2ym
b2yn.dat	formatted	b2yn
b2yp.dat	formatted	b2yp
b2yq.dat	formatted	b2yq
b2yr.dat	formatted	b2yr
b2yt.dat	formatted	b2yt
stra.dat	formatted	b2ar
weis.dat	formatted	b2ar

filename	formatted?	written by
b2ag.prt	formatted	b2ag
b2ah.prt	formatted	b2ah
b2ai.prt	formatted	b2ai
b2ar.prt	formatted	b2ar
b2co.prt	formatted	b2co
b2fu.prt	formatted	b2fu
b2mn.prt	formatted	b2mn
b2uf.prt	formatted	b2uf
b2yg.prt	formatted	b2yg
b2yh.prt	formatted	b2yh
b2yi.prt	formatted	b2yi
b2ym.prt	formatted	b2ym
b2yn.prt	formatted	b2yn
b2yp.prt	formatted	b2yp
b2yq.prt	formatted	b2yq
b2yr.prt	formatted	b2yr
b2yt.prt	formatted	b2yt

filename	formatted?	written by	read by	description
b2fgmtry	un*formatted	b2ag	b2co b2fu b2mn b2pl b2ts b2uf b2xd b2yg b2yp	geometry
b2fmovie	un*formatted	b2mn	b2ym	
b2fparam	un*formatted	b2mn	b2fu b2pl b2ts b2uf b2xd b2yh b2yp	
b2fpardf	un*formatted	b2ah	b2co b2mn	
b2fplasma	unformatted	b2fu b2mn	b2pl b2ts b2uf b2xd	plasma state used for plotting
b2fplasmf	formatted	b2uf	b2fu	plasma state used for plotting
b2frates	un*formatted	b2ar	b2mn b2yr	atomic physics rates
b2fstate	un*formatted	b2co b2mn	b2fu b2pl b2ts b2uf b2xd b2yi b2yi_gnuplot b2yp b2yt	final plasma state
b2fstati	un*formatted	b2ai	b2co b2mn	initial plasma state
b2fstatt	un*formatted	b2yt	plasma state on a new grid	
b2ftrace	un*formatted	b2mn	b2yq	residuals
b2ftrack	un*formatted	b2mn	b2yn	residuals
b2specp	formatted	b2yp		
b2tran	formatted	b2mn		

```

*dimens          (nx, ny; free format)
  96  24   96  24
*param           (param(0:99); free format)
  -1.0, 0.0, 0.75, 1.0, -0.010, 0.0, 0.040,
  5.0, 1.0, 0.02, 0.05, 0.05, 0.05,
  0.06, 1.0,
  100*0.0
'b2agfs_geometry'      'upgrade.dg-ascii.550'
  
```

***dimens** specifies the size of the grid

- first pair is nx & ny of the grid you want to run on
- second pair the size of the grid that was originally created
 - each needs to be an integer multiple of the corresponding entry of the first pair

***param** at least 100 additional numbers, of which

only the first is relevant for us

- 1.0** read the mesh data using the “simplified” format
- 2.0** read the mesh data using the “SONNET” format

b2agfs_geometry specifies the mesh file to be read

- searched for in (in order)
 1. the b2ag.exe.dir sub-directory of the current run directory
 2. the current run directory
 3. \$SOLPSTOP/data/meshes
- ... other options can be used to modify the mesh and magnetic field information that are read in from mesh file

```
*tlohi          (tlo, thi; real, free format)
  1.0e-1  1.0e4
*nlohi          (nlo, nhi; real, free format)
  1.0e18  1.0e22
*numnuc         (nnuc; integer, free format)
  3
*nucspec        (nz, izlo, izhi; integers, one triple per line)
  1  0  1
  6  0  6
  2  0  2
*flag           (either 'adpak' or 'strahl', free format)
  'strahl'
*tailep         (real, free format)
  0.08
```

- Discussed separately

b2ai: preparing the initial plasma state

```

*dimens          (ns; free format)
  12
*label           (lblps, character*60; free format)
  'Flat initial state for H + C + He, all charge states'
*specs           za      zn      am      za2
  '0'           0.0      1.0      2.0      0.0
  '1'           1.0      1.0      2.0      1.0
  '2'           0.0      6.0     12.011     0.0
  '3'           1.0      6.0     12.011     1.0
  '4'           2.0      6.0     12.011     4.0
  '5'           3.0      6.0     12.011     9.0
  '6'           4.0      6.0     12.011    16.0
  '7'           5.0      6.0     12.011    25.0
  '8'           6.0      6.0     12.011    36.0
  '9'           0.0      2.0      4.003     0.0
  '10'          1.0      2.0      4.003     1.0
  '11'          2.0      2.0      4.003     4.0
*naini           (na0(0:ns-1); free format)
  1.0e12  1.0e20  6*1e16  3*6e16  100*1.0e0
*ttini           (tt0; free format)
  1.0e2

```

***dimens** the number of charge states

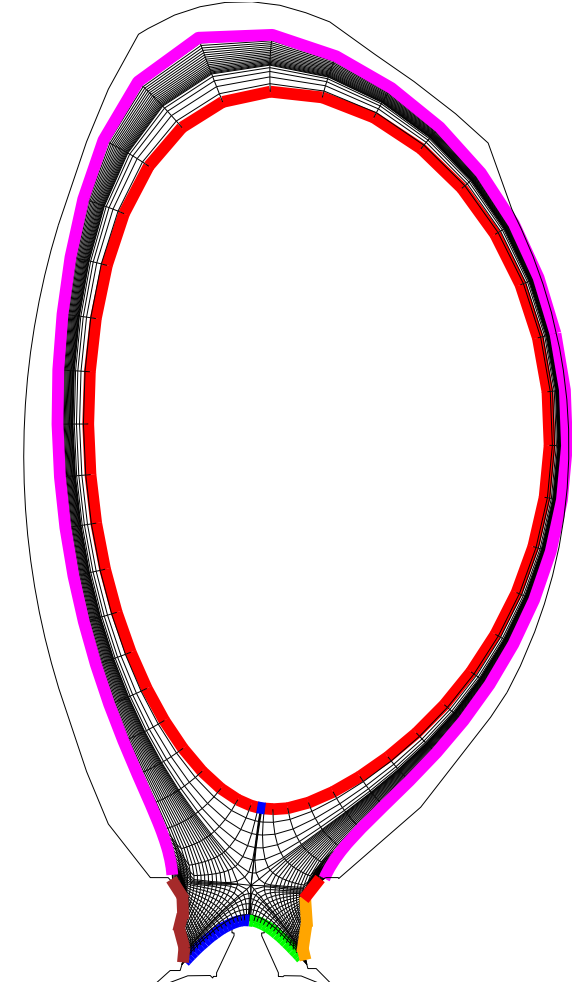
***label** a label

***specs** atomic charge, nuclear charge, atomic mass
and atomic charge squared for each charge state

***naini** initial densities for each of the charge states

***ttini** initial temperature

- the initial plasma has a flat density and temperature
- it is almost always faster to start from a plasma from a previous run



The double null mapping is even more complicated! See Xavier's talk.

b2ah: preparing the default physics parameters

This is for a D only case — not the same as for previous examples!

This file specifies the boundary conditions and transport coefficients for the default physics model.

The recommended way of specifying the actual values used for boundary conditions and transport coefficients is specified later.

```
*dimens          (ns; free format)
  2
*label           (lblps, character*60; free format)
  'Species D, standard SN'
*b2cmpa          basic parameters
*specs           za      zn      am      za2
  '0'           0.0      1.0      2.0      0.0
  '1'           1.0      1.0      2.0      1.0
```

***dimens** the number of charge states (here 2 corresponding to neutral and ionised D)

***label** a label

***b2cmpa** identifies a block of parameters, here “basic parameters”

***specs** atomic charge, nuclear charge, atomic mass and atomic charge squared

*b2cmpb	boundary conditions			
*cbregs				
'south'	3	0.25	0.75	2.00
'north'	1	2.0		
'west'	1	2.0		
'east'	1	2.0		

- *b2cmpb another basic block, for boundary conditions
- *cbregs specifies the regions where boundary conditions will be specified
 - the “south” boundary is broken into three pieces with one quarter of the cells in the first region, half of the cells in the second, and the remaining quarter in the third region (inner target private flux, core, outer private flux)
 - the “north”, “west” and “east” are not broken up, and correspond to the SOL boundary, inner target and outer target, respectively

Next comes one block for each of the regions identified:

*region

0 'south, private flux'

	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
*cbsna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	-0.01	0	0	0	0	0
'1'	0	0	-1.0e-03	0	0	0	0	0
*cbsmo	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	-1.0e+10	0	0	-1.0	0
'1'	0	0	0	0	0	0	-1.0	0
*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.00e-18	-100.0	-1.0e-01	0	0	0	0	-1.5
'1'	1.00e-18	-100.0	-1.0e-02	0	0	0	0	-1.5
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.00e-18	-100.0	-1.0e-04	0	0	0	0	-1.5
*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	1.0	0.3	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0


```
*region
1 'south, core plasma'
*cbsna      (0)      (1)      (2)      (3)      (4)      (5)      (6)      (7)
'0'         0      -1.0e+10    0        0        0        0        0        0
'1'    1.00e+29 -1.0e+10    0        0        0        0        0        0
*cbsmo      (0)      (1)      (2)      (3)      (4)      (5)      (6)      (7)
'0'         0        0        0      -1.0e+10    0        0        0        0
'1'         0        0        0      -1.0e+10    0        0        0        0
```

***region** region identifier

- here, the “south, core boundary”

***cbsna** boundary conditions for density

- two lines, one for each charge state

'0' a zero density boundary condition for D^0

'1' a density of $1 \times 10^{19} m^{-3}$ for D^+

***cbsmo** boundary conditions for momentum

'0' zero velocity

'1' zero velocity

The exact form of the specification is somewhat arcane: users are referred to the code for the definitive meaning, or to the write-up from one of Kuhn's group.

Users are encouraged to use the “physics” formulation of the boundary conditions specified later, rather than the “Bas” formulation listed here.

*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	8.00e+13	-1.0e+30	0	0	0	0	0	0
'1'	8.00e+13	-1.0e+30	0	0	0	0	0	0
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	8.00e+13	-1.0e+30	0	0	0	0	0	0

- ***cbshi** ion/neutral (or “atomic”) temperature/energy boundary condition
- one for each charge state, here both set to $500eV$ ($8 \times 10^{13}/1.6^{-19}/1 \times 10^{30}$)
- ***cbshe** the electron temperature/heat flux boundary condition
- here, also $500eV$

The boundary condition implemented arises from numerical combinations of the seven parameters with various quantities from the code.

*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

- *cbsch boundary condition for the electric potential equation
- *cbrec recycling coefficient
- *cbmsa [unused]
- *cbmsb [unused]

*region

2 'south, private flux'

	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
*cbsna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	-0.01	0	0	0	0	0
'1'	0	0	-1.0e-03	0	0	0	0	0
*cbsmo	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	-1.0e+10	0	0	-1.0	0
'1'	0	0	0	0	0	0	-1.0	0
*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.00e-18	-100.0	-1.0e-01	0	0	0	0	-1.5
'1'	1.00e-18	-100.0	-1.0e-02	0	0	0	0	-1.5
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.00e-18	-100.0	-1.0e-04	0	0	0	0	-1.5
*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	1.0	0.3	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

*region

3 'north, SOL/DIV outer wall'

	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
*cbsna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	-1.0e-03	0	0	0	0	0
*cbsmo	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	-1.0e+10	0	0	-1.0	0
'1'	0	0	0	0	0	0	-1.0	0
*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.00e-18	-100.0	-1.0e-01	0	0	0	0	-1.5
'1'	1.00e-18	-100.0	-1.0e-02	0	0	0	0	-1.5
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.00e-18	-100.0	-1.0e-04	0	0	0	0	-1.5
*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	1.0	0.3	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

*region

4 'west, target plate'

	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
*cbsna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	-1.0
'1'	0	0	0	0	0	0	0	-1.0
*cbsmo	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	-1.0e+10	0	0	0
'1'	0	0	1.0e+10	0	-1.0e+10	0	0	0
*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.00e-18	-100.0	0	0	0	0	0	-1.5
'1'	1.00e-18	-100.0	0	-0.1	0	0	0	-1.5
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.00e-18	-100.0	0	-0.1	0	0	0	-1.5
*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	1.0	0.5
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	1.00	0.3	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

*region								
5 'east, target plate'								
*cbsna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	-1.0
'1'	0	0	0	0	0	0	0	-1.0
*cbsmo	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	-1.0e+10	0	0	0
'1'	0	0	1.0e+10	0	-1.0e+10	0	0	0
*cbshi	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.00e-18	-100.0	0	0	0	0	0	-1.5
'1'	1.00e-18	-100.0	0	-0.1	0	0	0	-1.5
*cbshe	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.00e-18	-100.0	0	-0.1	0	0	0	-1.5
*cbsch	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	1.0	0.5
*cbrec	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	1.00	0.3	0	0	0	0	0	0
*cbmsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0	0	0	0	0	0	0	0
*cbmsb	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

*b2cmpt	transport coefficients							
*cfd0	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	1.0e+02	0	1.0	5.0e-20	1.5e-18	0	0	0
'1'	0.5	0	0	0	0	0	0	0
*cfdna	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	1.0	0	0	0	0	0
*cfdpa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	1.0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0

***b2cmpt** another basic block

***cfd0** auxiliary transport quantity

- $df0 = cfd0_0 + cfd0_1/(n_i/1.0e20) + cfd0_2 * \sqrt{T_i/(am * m_p)}/(cfd0_3 * n_i + cfd0_4 * n_e)$
- '0' 100 + 1 * $\sqrt{T_i/2m_p}/(5 \times 10^{-20} * n_i + 1.5 \times 10^{-18} * n_e)$
- '1' 0.5

***cfdna** density driven particle diffusion coefficient

- '0' 0
- '1' 1 * 0.5

***cfdpa** pressure driven particle diffusion coefficient

- '0' 1 * df0
- '1' 0

*cfvla	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	0	0	0	0	0	0
'1'	0	0	0	0	0	0	0	0
*cfvsa	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	1.0	0	0	0	0	0
'1'	0	0	0.4	0	0	0	0	0
*cfhci	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'0'	0	0	1.4	0	0	0	0	0
'1'	0	0	1.4	0	0	0	0	0
*cfhce	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	0.7	0	0	0	0	0	0	0

***cfvla** anomalous “pinch” velocity

'0' 0

'1' 0

***cfvsa** anomalous transverse momentum diffusivity

'0' 1 * df0

'1' 0.4 * 0.5

***cfhci** anomalous ion/neutral heat diffusivity

'0' 1.4 * df0

'1' 1.4 * 0.5

***cfhce** anomalous electron heat diffusivity

'-1' 0.7

*cfsig	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.0e-03	0	0	0	0	0	0	0
*cfalf	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)
'-1'	1.0e-03	0	0	0	0	0	0	0
*cflim	cflme	cflmi	cflmv	...				
'-1'	0.15	0.15	0.5	0	0	0	0	0
*endphy								

***cfsig** diffusivity corresponding to the anomalous electrical conductivity

***cfalf** diffusivity corresponding to the anomalous thermo-electric coefficient

***cflim** coefficients for flux limiting

***endphy** end of the data

Fuller descriptions are found in "b2cdcv.F" in the "documentation" sub-directory of the B2 source code.

```

*label          (lblmn: character*60)
  'D+C+He'
*b2cmpa  basic parameters
*b2cmpb  boundary conditions
*b2cmpt  transport coefficients
*cflim   cflme   cflmi   cflmv   ...
  '-1'    0.3    10.0    0.5     0     0     0     0     0
*endphy

```

***label** a label

***b2cmpa** basic parameters block

***b2cmpb** boundary conditions block

***b2cmpt** transport coefficients block

***cflim** override the flux limiters from b2ah

***endphy** end of block

- any of the parameters from b2ah can be altered

b2mn: the main program, continued

'b2mndr_ntim'	'1000'	'b2mndr_na0eps'	'1e14'
'b2mndr_dtim'	'1.0e-6'	'b2mndr_cpu'	'82800.0'
'b2mndt_nstg0'	'1'	'b2mndr_savecpu'	'0.0'
'b2mndt_nstg1'	'1'	'b2ux5p_style'	'2'
'b2mndt_nstg2'	'10'	'b2ux5p_nltrsol'	'2'
'b2mndt_rxf'	'0.5'	'b2stbc_sna0ep'	'1.0e-10'
'b2mndr_mvinc'	'1'	'b2trcl_lluciani'	'1'
'b2mndr_mvnum'	'0'	'b2trcl_lvis21'	'1'
'b2mndr_b2time'	'1'	'b2trcl_lthf21'	'1'

b2mndr_ntim do 1000 iterations

b2mndr_dtim with a time-step of $1 \times 10^{-6} s$

b2mndt_nstg2 with 10 internal iterations per time-step

b2mndr_cpu do no more than 82800 cpu-seconds (23 cpu-hours) of work

b2ux5p_style use the sparse matrix solver

b2trcl_lluciani implement the Luciani parallel flux limiter

b2trcl_lvis21 use the 21-moment model for parallel viscosity

b2trcl_lthf21 use the 21-moment model for the parallel thermal force

b2mn: the main program, continued

```
'b2news_potit'      '100'
'b2news_potok'      '1e-2'
'b2sihs_phm3'       '1.0'
'b2stbr_sput_frc'   '0.0000'
'b2stbc_boundary_namelist' '1'
'b2srdt_numerics_namelist' '1'
'b2tqna_transport_namelist' '1'
'b2tqna_new_df0'    '0'
'b2mndr_eirene'     '1'
'b2mndr_rescale_neutrals' '1e-0'
'b2mndr_rescale_neutrals_sources' '1e-10'
'b2mwti_jxi'       38
'b2mwti_jxa'       56
'b2news_recalculate_contributions' '0'
```

b2news_potit do no more than 100 iterations to solve the potential equation

b2news_potok solve the potential equation to $0.01V$

b2stbc_boundary_namelist read the “physics” style boundary coefficients from `b2.boundary.namelist`

b2srdt_numerics_namelist read some numerics related data from `b2.numerics.namelist`

b2tqna_transport_namelist read the “physics” style transport coefficients from `b2.transport.namelist`

b2mndr_eirene use eirene (for a stand-alone run this would be 0)

b2mndr_rescale_neutrals don't rescale the neutrals (set to $1e-10$ when switching from the fluid model to eirene)

b2mndr_rescale_neutrals_sources rescale the neutral sources within B2 by 1×10^{-10}

b2mn: b2.transport.parameters

```
&transport
flag_dna=1, parm_dna=12*1,
flag_dpa=0, parm_dpa=12*0,
flag_vla=0, parm_vla=12*0,
flag_vsa=0, parm_vsa=12*0,
flag_hci=1, parm_hci=12*2,
flag_hce=1, parm_hce=2,
flag_sig=0, parm_sig=0,
flag_alf=0, parm_alf=0,
/
```

Model's used

- 0** use Bas' model
- 1** constant
- 2** $1/n_e$
- 3** Bohm
- 4** flux-scaled

dna set the density driven transport to model 1 with value 1.0

dpa set the pressure driven transport to model 0 (i.e. use b2ah)

vla use b2ah

vsa use b2ah

hci set the ion thermal diffusivity using model 1 with value 2

hce set the electron thermal diffusivity using model 1 with value 2

sig use b2ah

alf use b2ah

```
&numerics
```

```
dtco(0,1)= 12*100.0, dtco(0,2)=12*1.0, dtco(0,3)=12*1.0, dtco(0,4)=12*1.0,  
dtmo(0,1)= 12*100.0, dtmo(0,2)=12*1.0, dtmo(0,3)=12*1.0, dtmo(0,4)=12*1.0,  
dtee(1)= 100.0, 1.0, 1.0, 1.0,  
dtei(1)= 100.0, 1.0, 1.0, 1.0,  
/
```

- re-scale the time-step for the core region (region 1) by 100
- for the continuity, momentum, electron and ion heat equations

&boundary

NBC=6,

BCCHAR=	'S',	'S',	'S',	'N',	'W',	'E',
BCPOS=	-1,	-1,	-1,	24,	-1,	96,
BCSTART=	24,	0,	72,	0,	-1,	-1,
BCEND=	71,	23,	95,	95,	24,	24,
BCENE=	8,	9,	9,	9,	3,	3,
BCENI=	8,	9,	9,	9,	3,	3,
BCPOT=	0,	0,	0,	0,	3,	3,

NBC 6 boundaries

BCCHAR type of boundary (north, east, south, west)

BCPOS position on that face

BCSTART start position of boundary

BCEND end position of boundary

BCENE type of boundary condition to be applied for
electron energy equation

BCENI type of boundary condition to be applied for
ion energy equation

BCPOT type of boundary condition to be applied for

potential equation

where

1 set a value

3 sheath

8 prescribed total flux with constant flux density

9 decay length


```

ENEPAR=      2e6,   0.01,   0.01,   0.01,   0.90,   0.90,   ENEPAR(2,2)=1e2, 1e2, 1e2,
ENIPAR=      2e6,   0.01,   0.01,   0.01,   2.50,   2.50,   ENIPAR(2,2)=1e2, 1e2, 1e2,
POTPAR=      0.00,   0.00,   0.00,   0.00,   0.00,   0.00,   POTPAR(1,2)=6*0,
BCCON(0,1)   =      8,      1,      8,      8,      8,      8,      8,      8,      8,      8,      8,      1,
  CONPAR(0,1,1) = 0e19, 4e19, 0e19, 0e19, 0e19, 0e19, 0e19, 0e19, 0e19, 0e19, 0e19, 1e16,
BCCON(0,2)   =     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,
  CONPAR(0,2,1) =-1e-2, -1e-3, -1e-2, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-2, -1e-3, -1e-3,
BCCON(0,3)   =     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,
  CONPAR(0,3,1) =-1e-2, -1e-3, -1e-2, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-2, -1e-3, -1e-3,
BCCON(0,4)   =     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,     10,
  CONPAR(0,4,1) =-1e-2, -1e-3, -1e-2, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-3, -1e-2, -1e-3, -1e-3,
BCCON(0,5)   =      5,      3,      5,      3,      3,      3,      3,      3,      3,      5,      3,      3,
  CONPAR(0,5,1) = 0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,
BCCON(0,6)   =      5,      3,      5,      3,      3,      3,      3,      3,      3,      5,      3,      3,
  CONPAR(0,6,1) = 0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,  0.00,

```

ENEPAR additional parameters for model

ENIPAR additional parameters for model

POTPAR additional parameters for model

BCCON choose which model to apply for the continuity boundary condition

CONPAR additional parameters for model

```

BCMOM(0,1)      =      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,
  MOMP(0,1,1) = 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, mompar(0,1,2)=12*1e5,
BCMOM(0,2)      =      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,
  MOMP(0,2,1) = 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, mompar(0,2,2)=12*1e5,
BCMOM(0,3)      =      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,
  MOMP(0,3,1) = 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, mompar(0,3,2)=12*1e5,
BCMOM(0,4)      =      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,      4,
  MOMP(0,4,1) = 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, mompar(0,4,2)=12*1e5,
BCMOM(0,5)      =      1,      3,      1,      3,      3,      3,      3,      3,      3,      1,      3,      3,
  MOMP(0,5,1) = 0.00, 1.00, 0.00, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00, 0.00, 1.00, 1.00, mompar(0,5,2)=12*0,
BCMOM(0,6)      =      1,      3,      1,      3,      3,      3,      3,      3,      3,      1,      3,      3,
  MOMP(0,6,1) = 0.00, 1.00, 0.00, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00, 0.00, 1.00, 1.00, mompar(0,6,2)=12*0,
GAMMAI=1.666666667,GAMMAE=0.5,
LBNDUSR=f,
/
  
```

BCMOM choose the boundary condition model to apply to the momentum equation

MOMP additional parameters for model